Lecture 10: SRAM and DRAM

CMPS 221 – Computer Organization and Design

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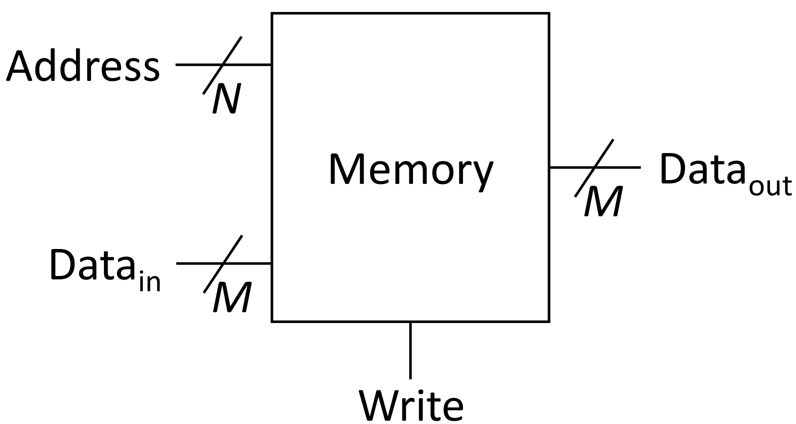
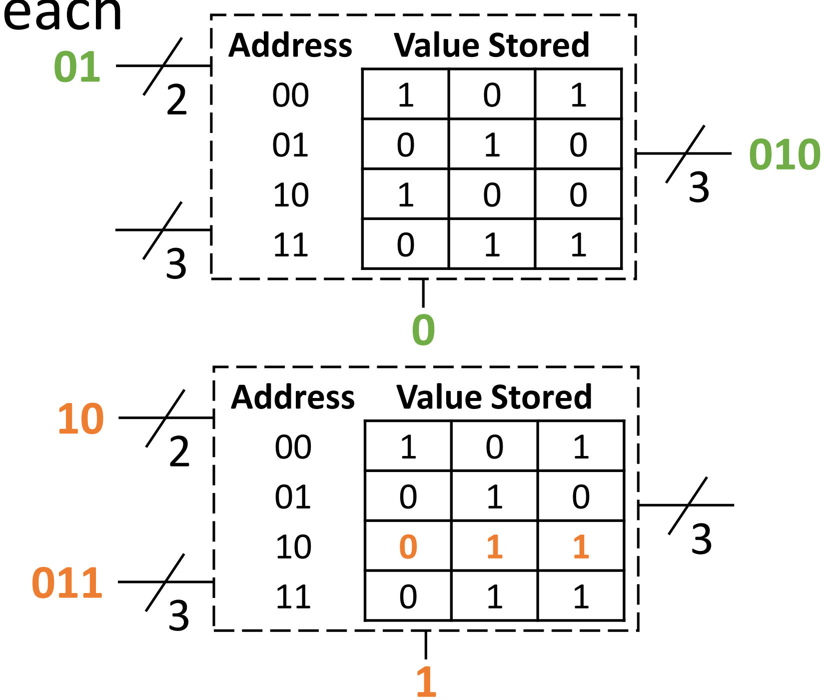
Recall: Memory

- A **memory** is a logic block that stores data
  - It takes a memory address as input and outputs the value stored at that address or writes to it
  - A memory with a *N*-bit address and M-bit output value stores 2^N entries with M bits each
  - Different technologies for building memories will be covered later





Types of Memory

- A **register file** (RF) is a type of memory
  - The register file we have built:
    - Can read two registers and write one register every clock cycle
    - May read and write to the same register on the same cycle
    - Typically has a small number of registers

- Another type of memory is **static random access memory** (SRAM)
  - Higher capacity than RF (larger and denser)
  - Tradeoffs:
    - Slower access latency than RF
    - Can either read or write during a cycle, not both

Register File vs. SRAM

- Registers may be read and written on the same cycle
  - Register value must be *held the entire cycle* and only updated at the end on the active edge
  - Need *flip-flops* to guarantee this property
- SRAM is either read or written on the same cycle, but not both
  - Can *update during a cycle*, not just on the active edge
    - If the output is being written then it is not being read (does not need to be held for the entire cycle)
  - *Latches* are sufficient (no need for flip-flops)

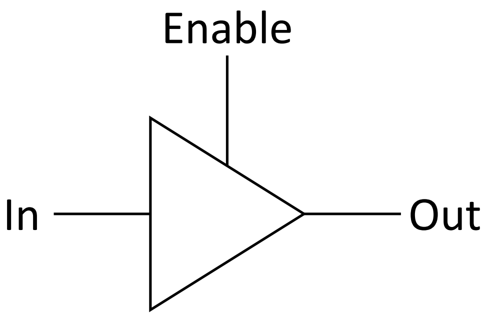
Register File vs. SRAM

- Register files typically have a <u>small</u> number of registers (e.g., 32 registers in MIPS)
  - Need a 32-to-1 mux to select the right register to read

- SRAM stores a <u>large</u> number of entries (e.g., 2M)
  - Building a 2M-to-1 mux is infeasible
  - Need a different technique to select the right entry to read

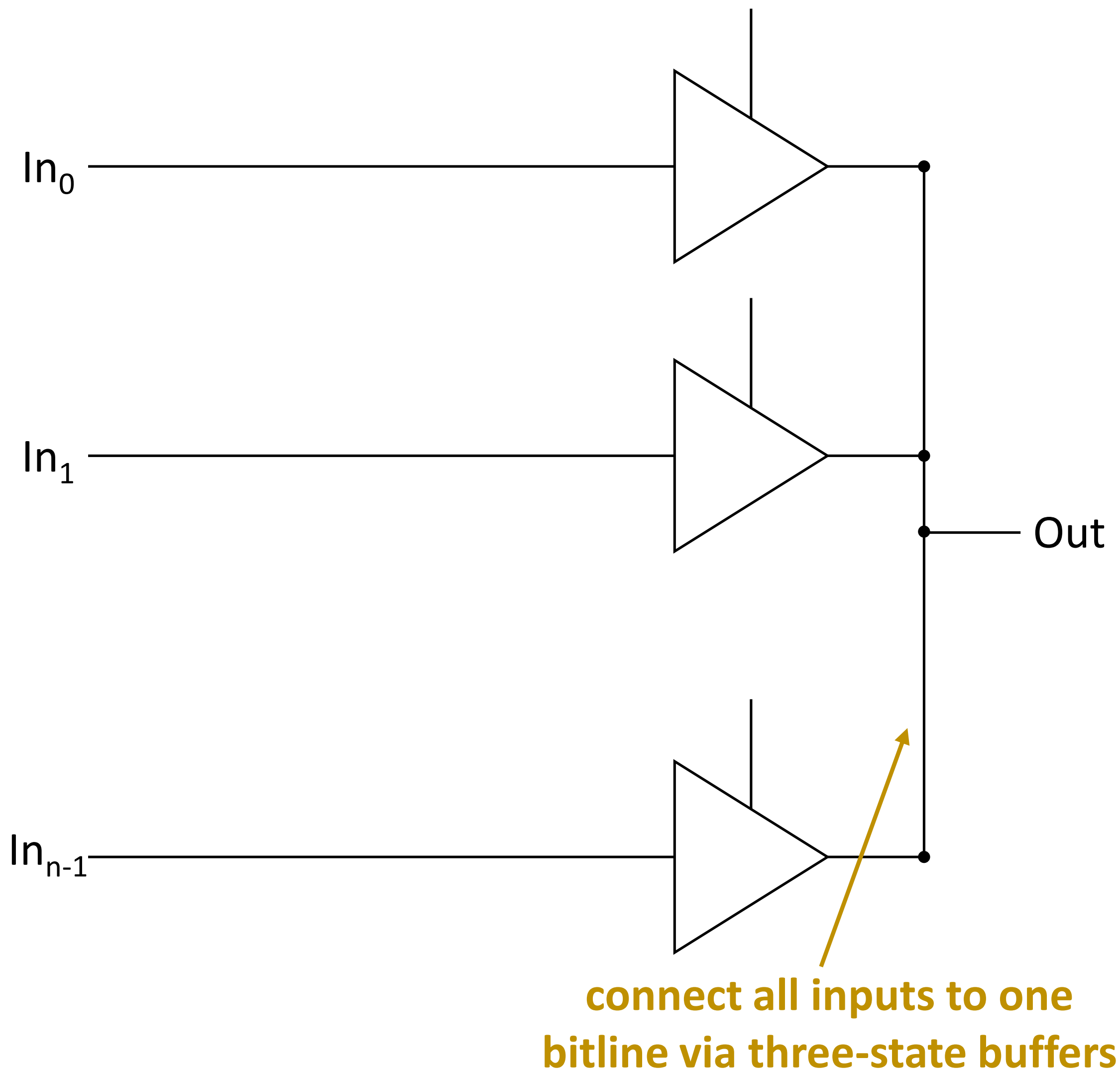
Three-State Buffers

- A **three-state buffer** is a circuit whose output can be in one of three states:
  - Low voltage (0)
  - High voltage (1)
  - High impedance (Z)
    - Open circuit that does not conduct electricity

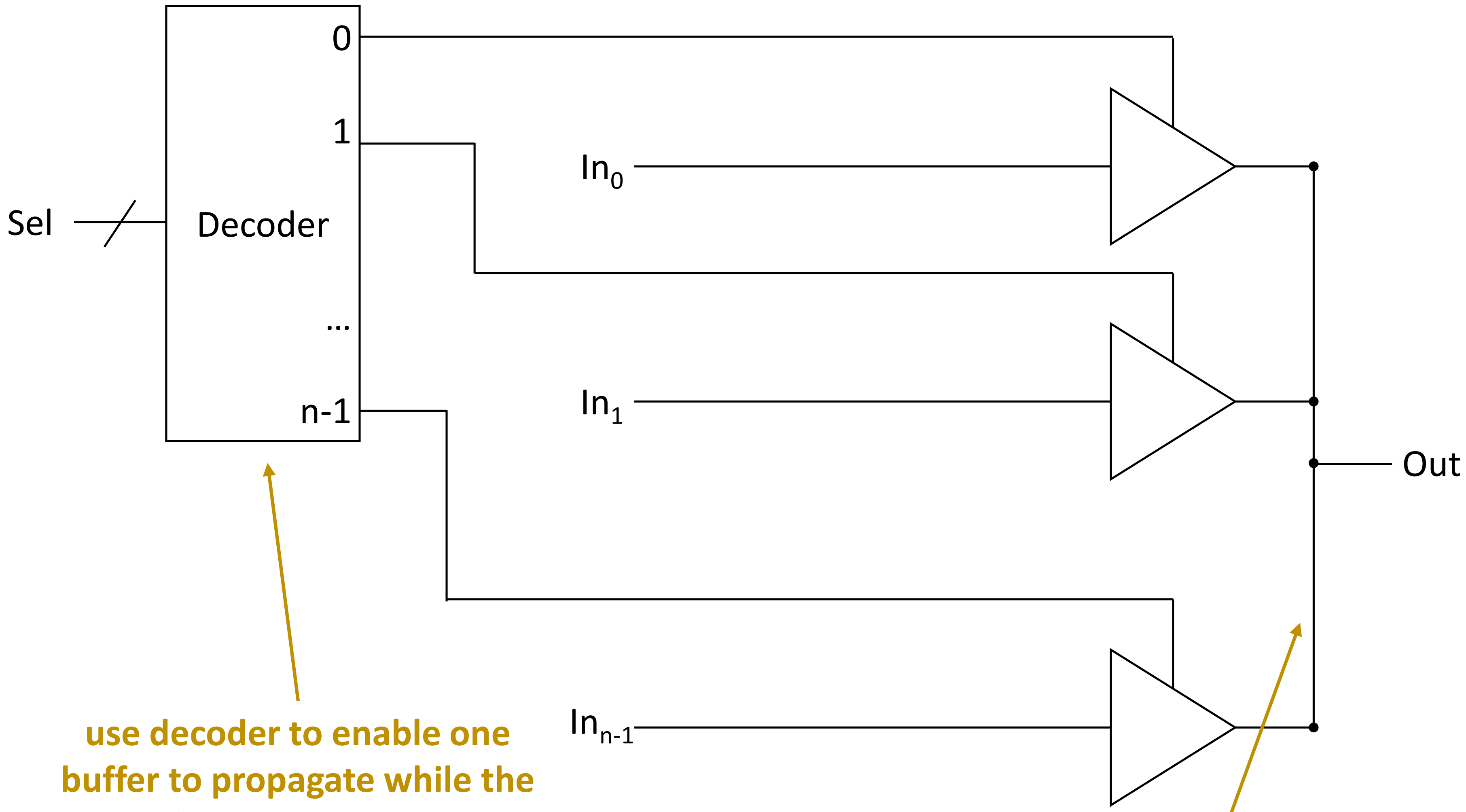


Enable	In	Out
0	0	Z
0	1	Z
1	0	0 (In)
1	1	1 (In)

Three-State Buffers for Selection



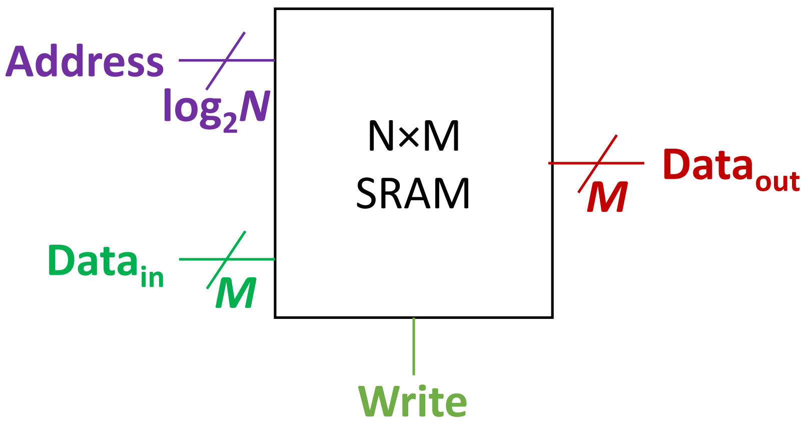
Three-State Buffers for Selection



use decoder to enable one buffer to propagate while the others remain open

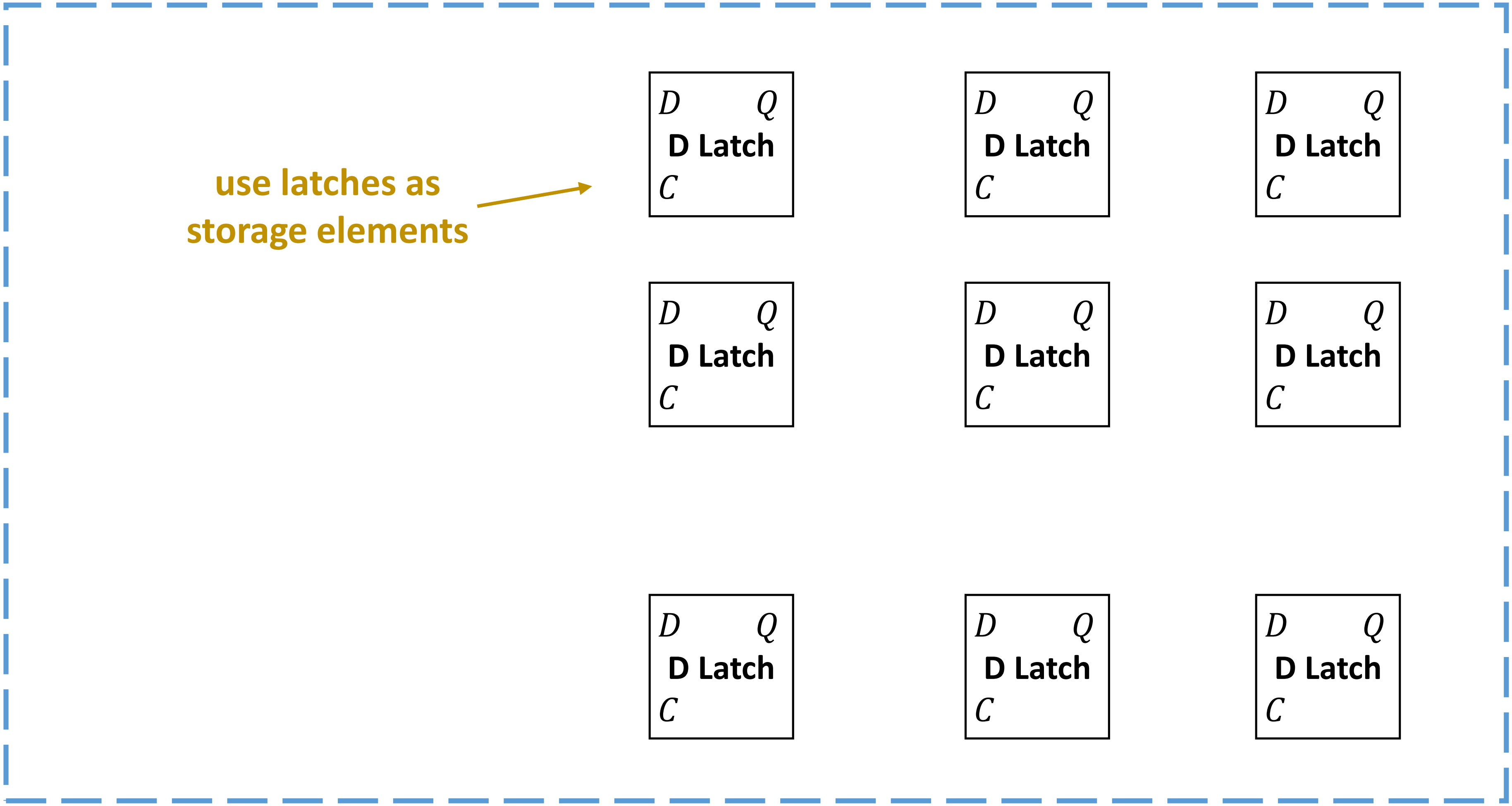
connect all inputs to one bitline via three-state buffers

Implementing SRAM

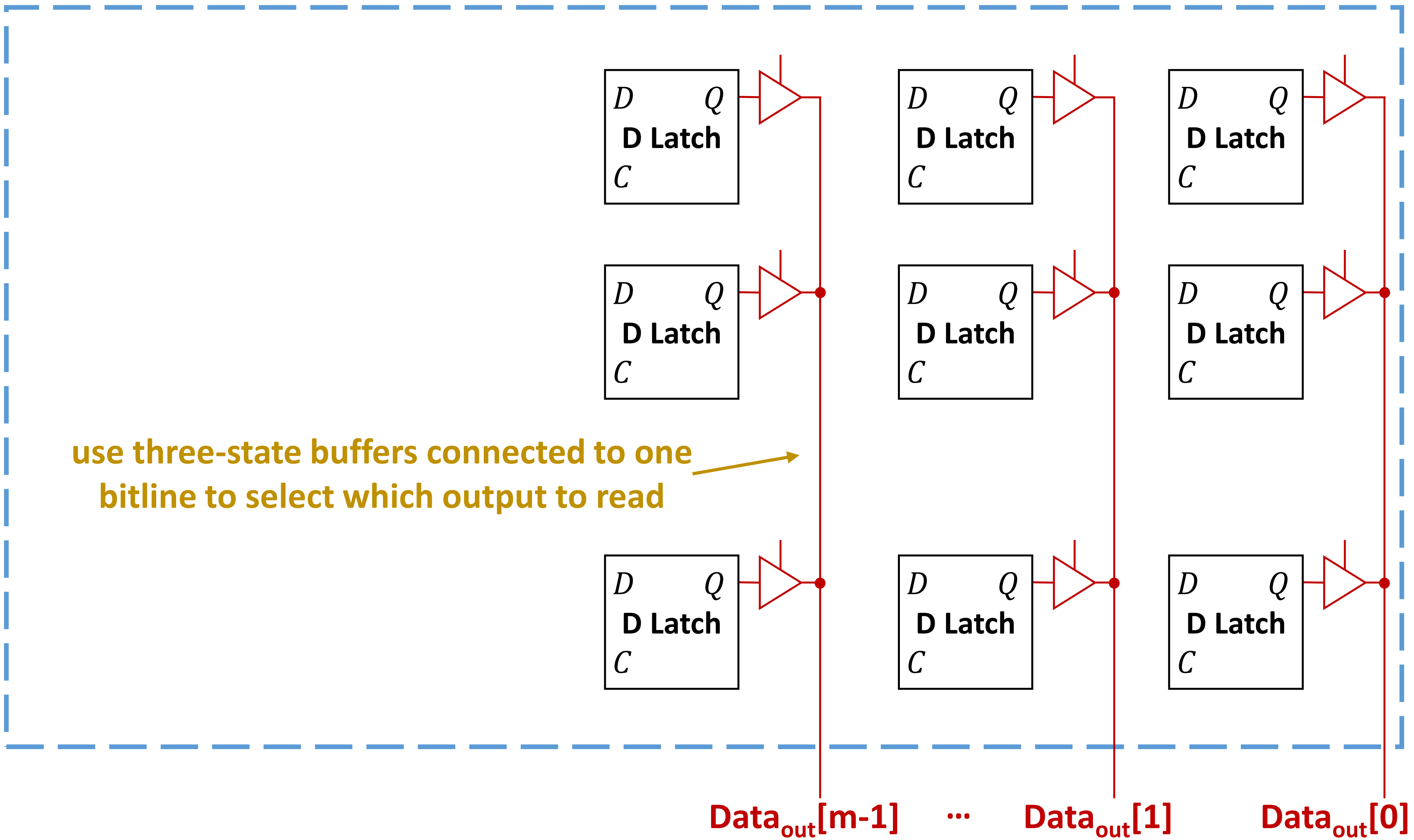


- Key points:
  - Use latches as 1-bit storage elements
  - Use three-state buffers to select which entry to read

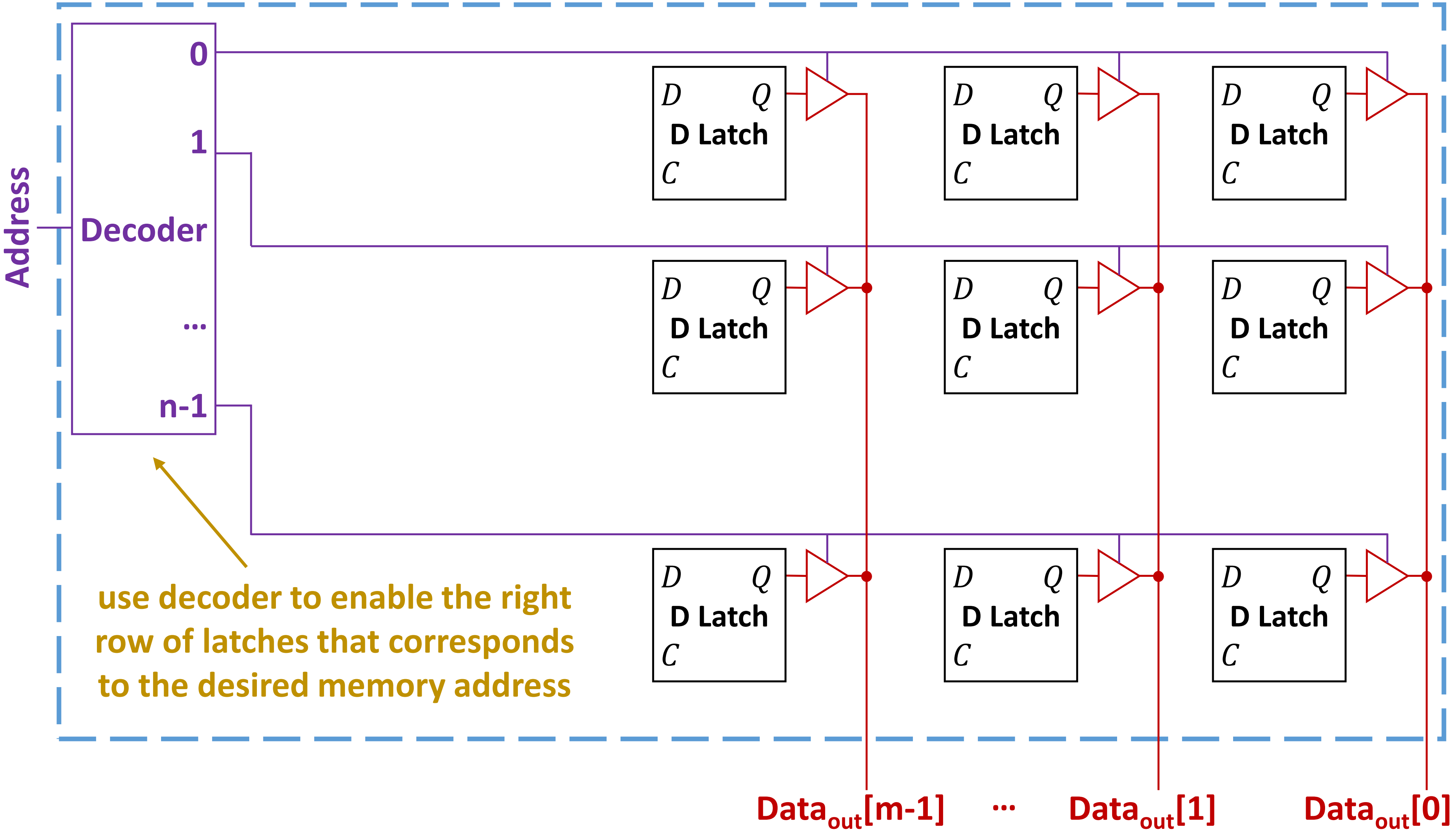
Implementing SRAM



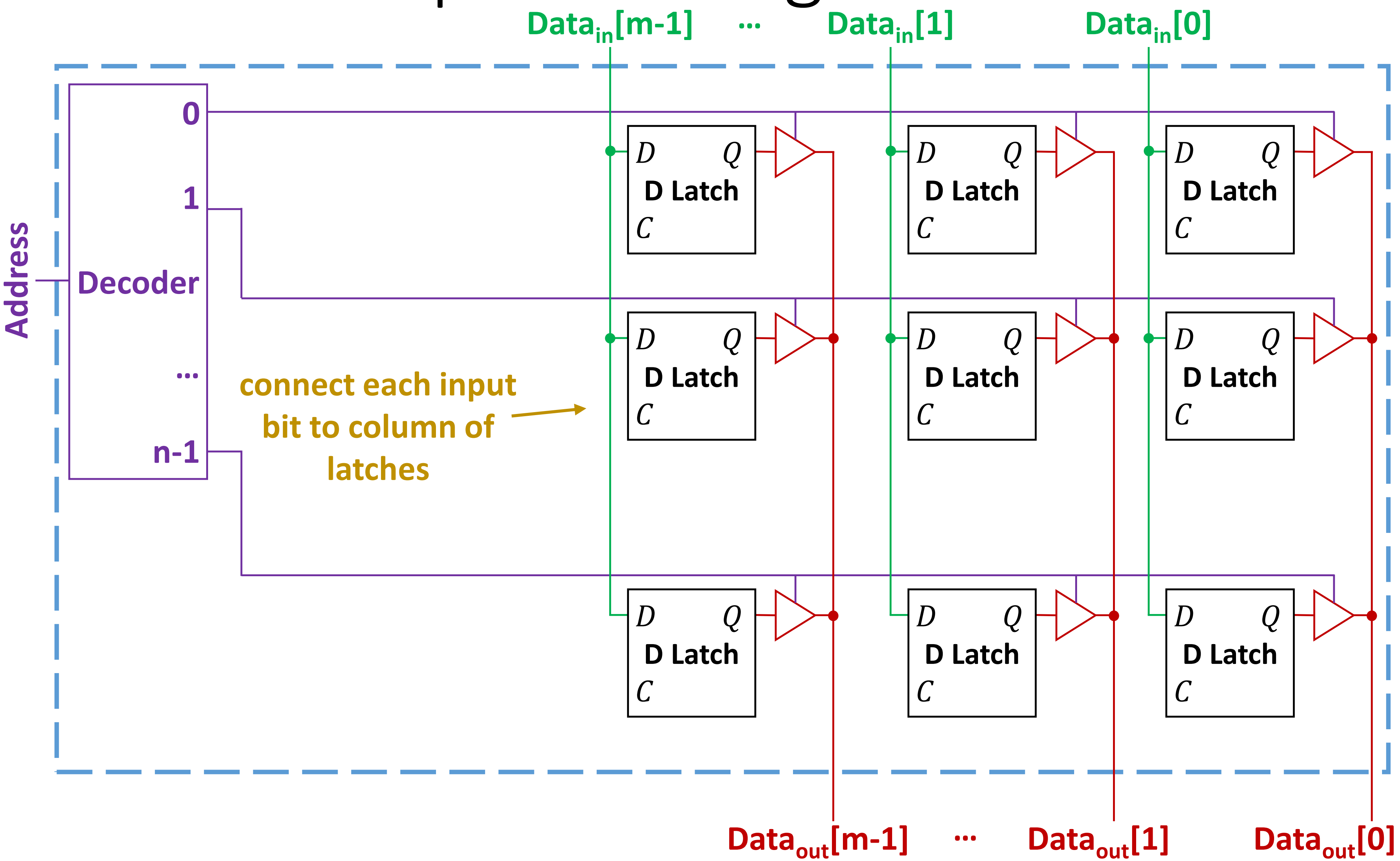
Implementing SRAM



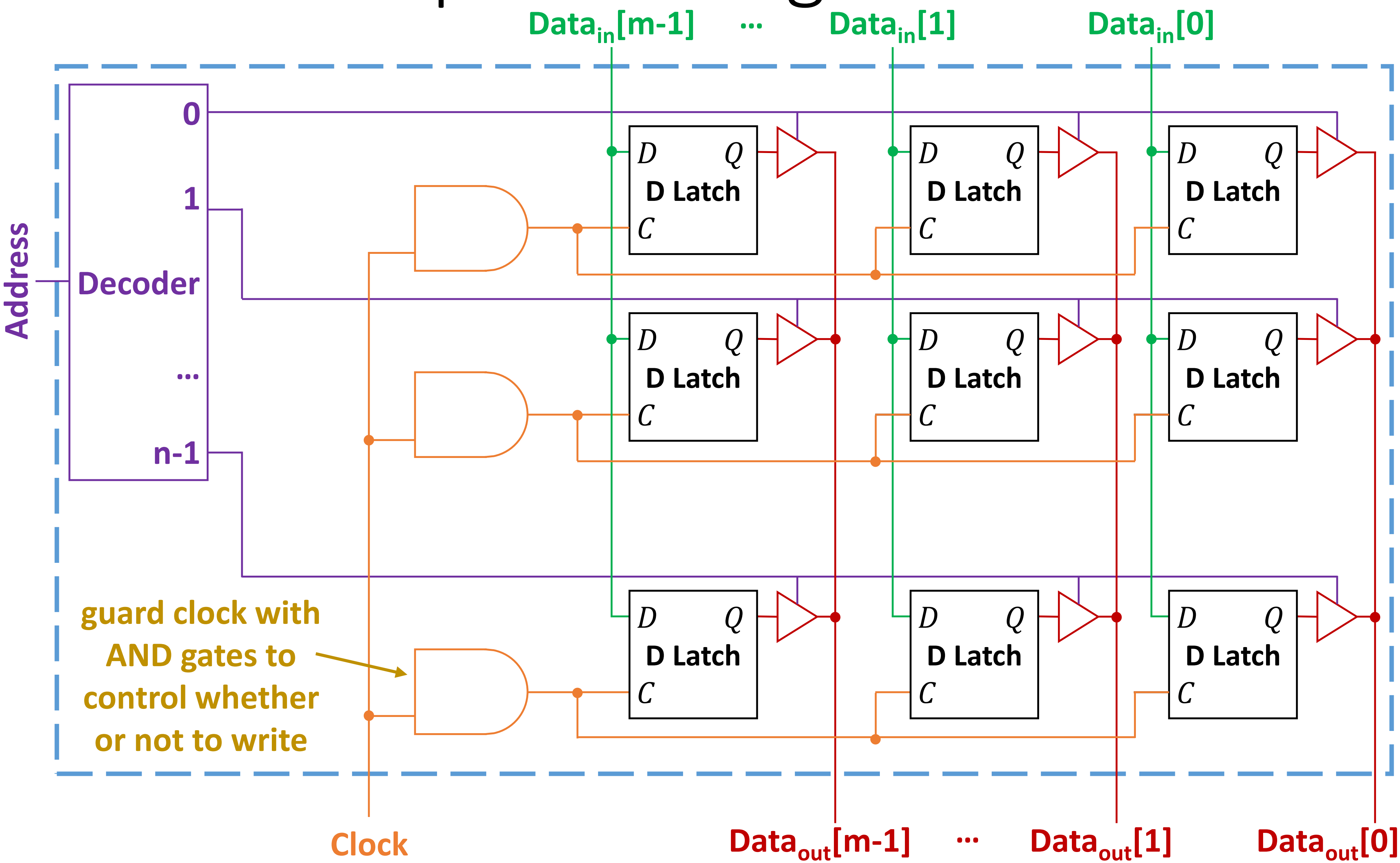
Implementing SRAM



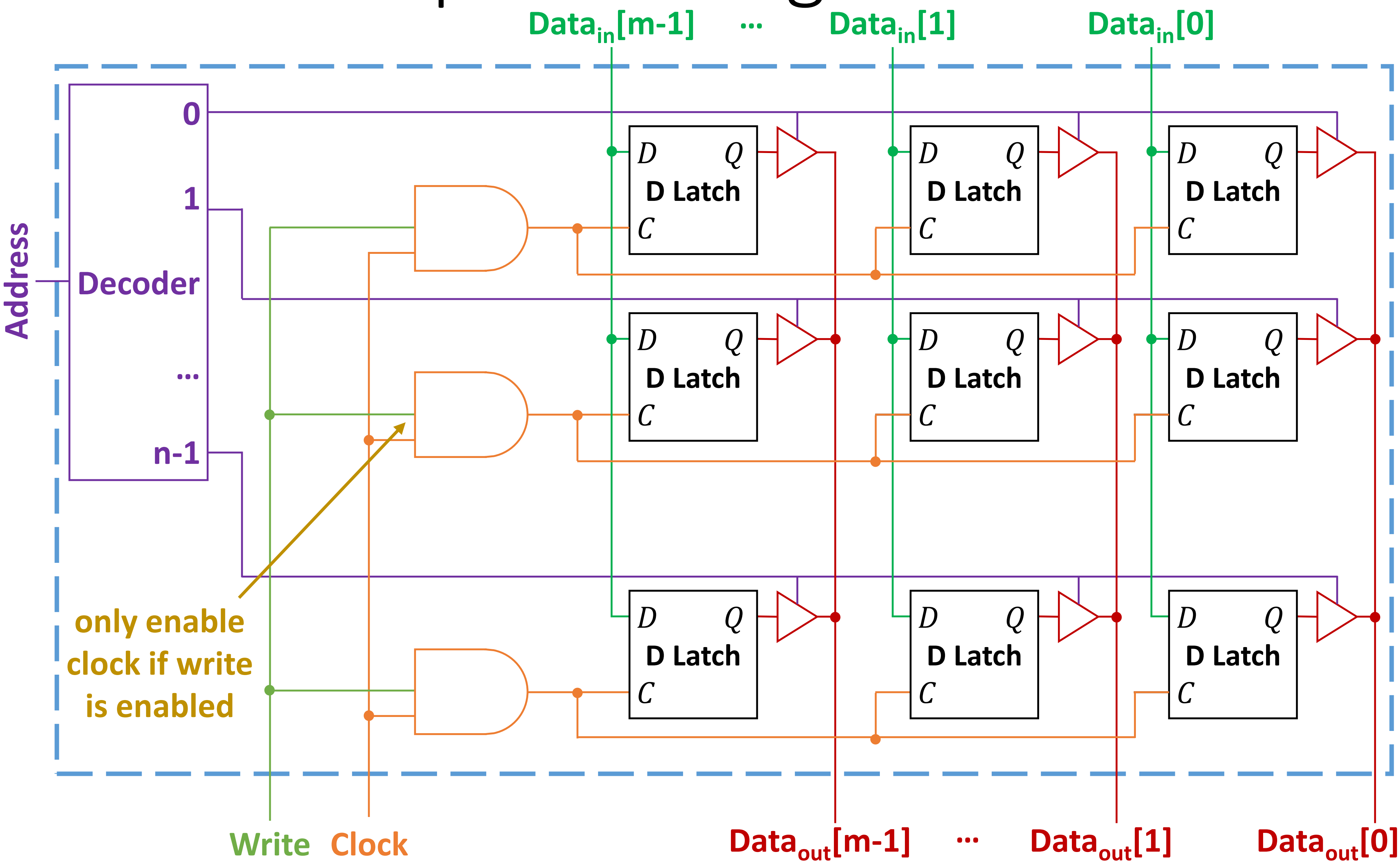
Implementing SRAM



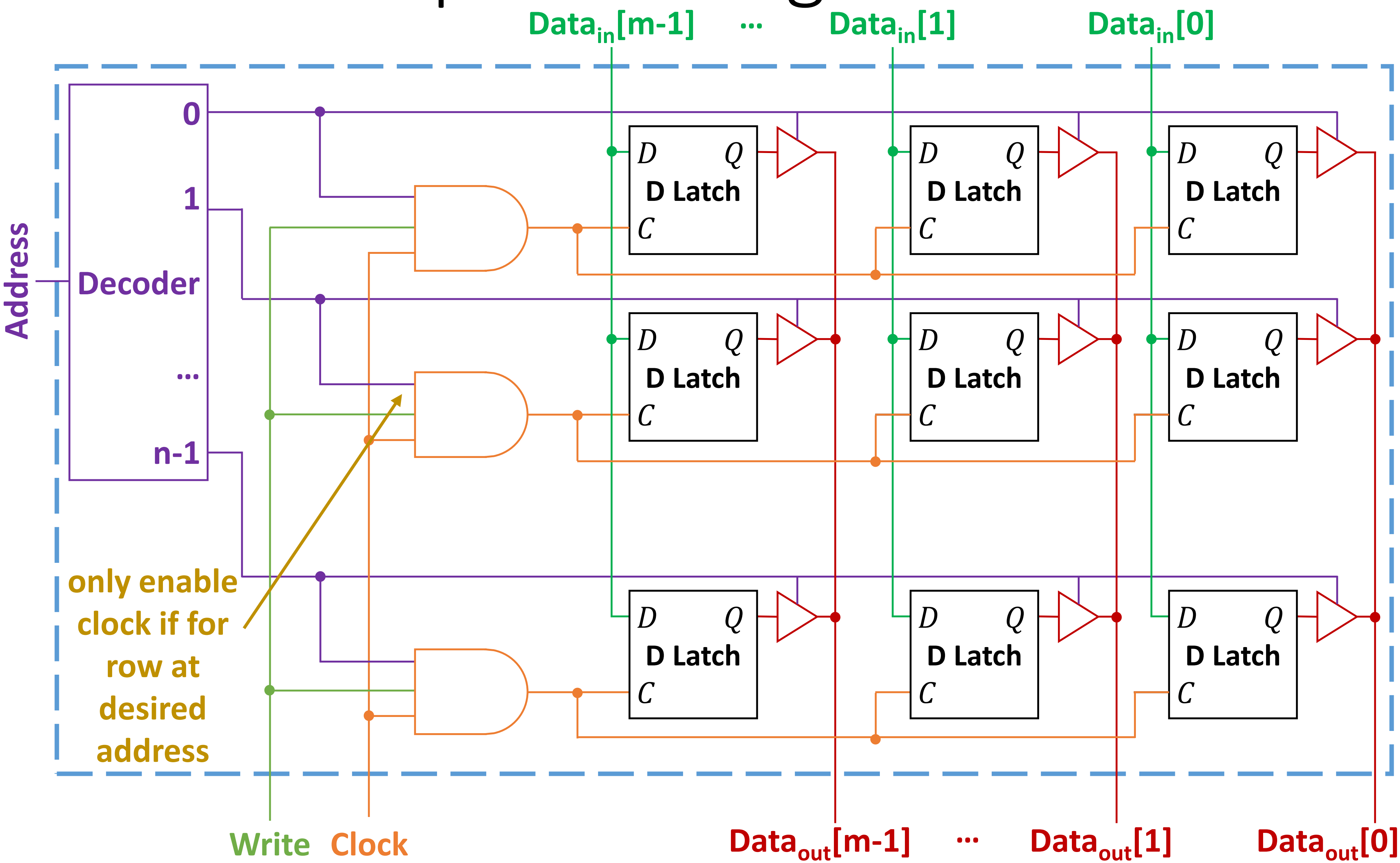
Implementing SRAM



Implementing SRAM



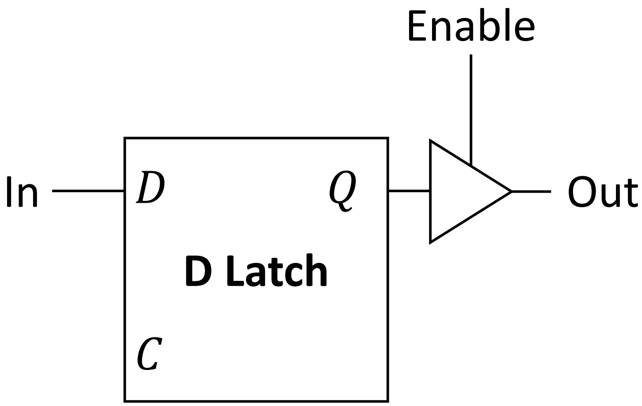
Implementing SRAM



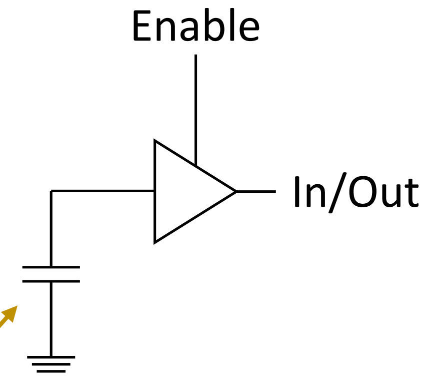
DRAM

- Another type of memory is **dynamic random access memory** (DRAM)
  - Can either read or write on a cycle (like SRAM, unlike RF)
  - Has higher capacity than SRAM (larger and denser)
  - Tradeoffs:
    - Has slower access latency than SRAM
    - *Dynamic*: data fades away with time and needs to be refreshed

SRAM vs. DRAM



SRAM 1-bit storage unit



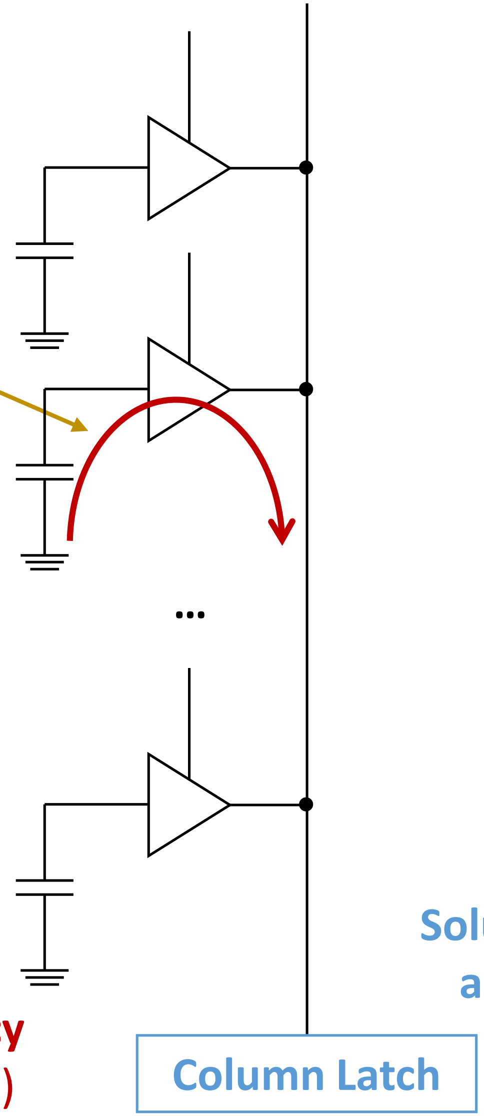
DRAM 1-bit storage unit

using a single capacitor instead of a D latch
(stores high charge if 1, no charge if 0)

Advantage: higher density
(capacitor needs less area than latch)

Destructive Reads

value is read by discharging the capacitor which destroys the value stored

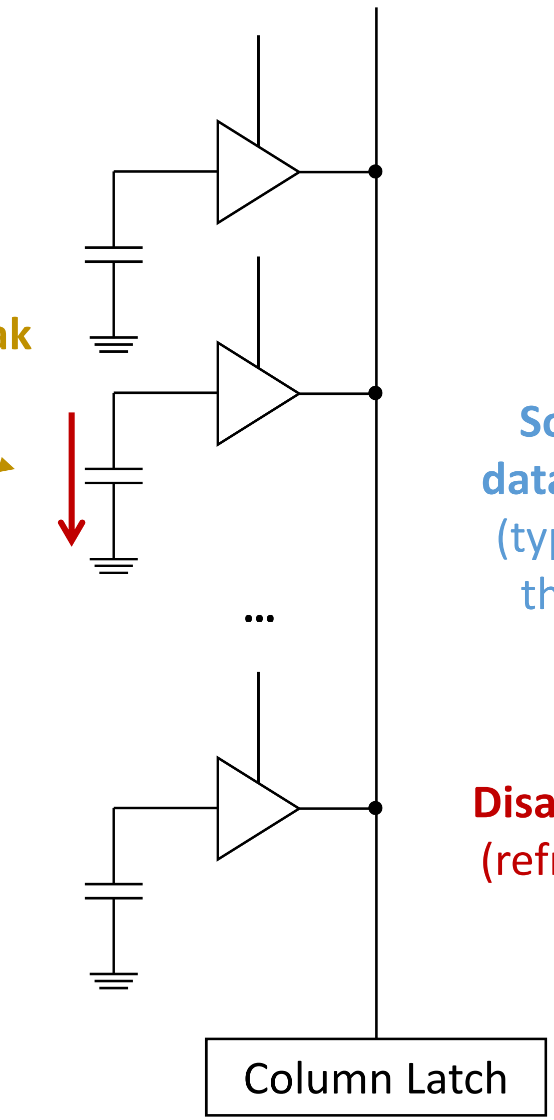


Solution: store read value in a latch and write it back after it is read

Disadvantage: slower read latency
(need to time to write value back)

Leakage

even if not read, the charge stored in the capacitor may leak



Solution: periodically "refresh" data by reading it and writing back
(typically, DRAM spends 1%-2% of the time on refresh operations)

Disadvantage: slower access latency
(refresh interferes with normal read and write operations)

Textbook Sections

- Some of the content in these slides corresponds to:
  - Textbook:
    - *Computer Organization and Design, 5th Edition by David Patterson and John Hennessy, Morgan Kaufmann, 2014.*
  - Sections:
    - B.9